

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409419
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Noguchi; Hiroshi et al.

Bubble generation device and liquid filtration device

Abstract

A bubble generation device capable of causing bubbles to hit filtration membranes almost simultaneously includes a housing having side plates and a top plate fixed to the side plates; bubble release ports disposed therein and arranged along a longitudinal direction of the top plate; and a bubble release chamber disposed in the housing and communicated with the bubble release ports. A folded-back path is formed by at least a folded-back plate and an inner plate disposed therein. One end and another end, respectively, of each of the folded-back plate and the inner plate in the longitudinal direction, are fixed to one of the two side plates and another of the two side plates, which face each other. One side and another side, respectively, of the folded-back path in the lateral direction are communicated with the bubble release chamber and a gas storage chamber.

Inventors:	Noguchi; Hiroshi (Singapore, SG), Niwa; Terutake (Singapore, SG), Nakagawa; Akitoshi (Tokyo, JP)
Applicant:	MEIDENSHA CORPORATION (Tokyo, JP)
Family ID:	87882186
Assignee:	MEIDENSHA CORPORATION (Tokyo, JP)
Appl. No.:	18/842719
Filed (or PCT Filed):	February 28, 2023
PCT No.:	PCT/JP2023/007262
PCT Pub. No.:	WO2023/167173
PCT Pub. Date:	September 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250108335 A1	Apr. 03, 2025

Foreign Application Priority Data

JP 2022-031598 Mar. 02, 2022

Publication Classification

Int. Cl.: B01D65/02 (20060101); B01D69/06 (20060101); C02F1/44 (20230101)

U.S. Cl.:

CPC B01D65/02 (20130101); B01D69/06 (20130101); C02F1/44 (20130101); B01D2321/185 (20130101); C02F2303/16 (20130101)

Field of Classification Search

CPC: B01D (65/02); B01D (69/06); B01D (2321/185); C02F (1/44); C02F (2303/16)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0056831	12/2016	Tanaka et al.	N/A	N/A
2017/0120197	12/2016	Tanaka et al.	N/A	N/A
2023/0321605	12/2022	Noguchi et al.	N/A	N/A
2023/0321614	12/2022	Noguchi et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
S63136728	12/1987	JP	N/A
2018079442	12/2017	JP	N/A
2020065976	12/2019	JP	N/A
WO-2015/146611	12/2014	WO	N/A
WO-2015/146686	12/2014	WO	N/A
WO-2022/059478	12/2021	WO	N/A
WO-2022/059479	12/2021	WO	N/A

Primary Examiner: Orme; Patrick

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a bubble generation device and a liquid filtration device using the same.

BACKGROUND ART

(2) Conventionally, in a liquid filtration device that filtrates a liquid using a filtration membrane, in

order to suppress clogging of the filtration membrane, it is common to remove contamination substances from the surface of the filtration membrane by scrubbing. Scrubbing is a method in which bubbles are released from below a filtration membrane toward the filtration membrane by a bubble generation device, and a liquid near a surface of the filtration membrane is vigorously shaken by the bubbles.

(3) A known bubble generation device includes a gas storage chamber and a folded-back path communicating with an upper portion thereof. The gas storage chamber stores a liquid therein and stores a gas supplied into the liquid above the liquid. The folded-back path communicates with the upper portion of the gas storage chamber and extends to the below, and is then folded back to extend to the above.

(4) As a bubble generation device having such a configuration, a bubble generation device described in Patent Literature 1 is known. The bubble generation device includes a gas storage path as a gas storage chamber and a gas guide path as a folded-back path. The gas storage path and the gas guide path are installed in the liquid. A gas is supplied from the outside into the liquid in a liquid storage path, floats in the liquid, and is stored in the upper portion of the liquid storage path. As the storage amount of the gas in the liquid storage path increases, the gas eventually enters the gas guide path from the upper portion of the liquid storage path. In the gas guide path, the gas moves to the below while pushing down the liquid in the gas guide path, and is directed toward the folded-back portion of the gas guide path. The gas that has reached the folded-back portion of the gas guide path is connected to the gas in the subsequent gas storage chamber, and floats at once while pushing up the liquid existing on the downstream side of the folded-back portion. Then, the gas is released as bubbles from the opening on the outlet side of the gas guide path.

CITATION LIST

Patent Literature

(5) Patent Literature 1: WO 2015/146686 A

SUMMARY OF INVENTION

Technical Problem

(6) In the bubble generation device described in Patent Literature 1, since the opening at the end of the pipe constituting the gas guide path serves as a bubble release port for releasing bubbles as it is, one spherical bubble is intermittently released. In the bubble generation device having such a configuration, in order to bring bubbles into contact with a wide range of the filtration membrane, it is necessary to dispose a plurality of gas guide paths side by side and release the bubbles generated from the gas guide paths toward different portions of the filtration membrane.

(7) On the other hand, scrubbing is a cleaning method in which contamination substances are removed from the surface of the filtration membrane by vibration of the filtration membrane due to contact with bubbles. Therefore, when a plurality of bubbles is released toward different portions of the filtration membrane, it is desirable to apply large vibration by hitting the bubbles against the filtration membrane almost simultaneously.

(8) However, in the bubble generation device described in Patent Literature 1, when a plurality of gas guide paths is arranged and bubbles are released from the gas guide paths, it is difficult to make gas supply speeds to the respective gas guide paths uniform. For this reason, the release cycles of the bubbles emitted from the plurality of gas guide paths cannot be synchronized, and it is difficult to hit the plurality of bubbles against the filtration membrane almost simultaneously.

(9) The present invention has been made in view of the above background, and an object thereof is to provide a bubble generation device capable of hitting a plurality of bubbles against a filtration membrane almost simultaneously, and a liquid filtration device using the same.

Solution to Problem

(10) An aspect of the present invention is a bubble generation device including a gas storage chamber that stores a liquid therein and stores a gas supplied into the liquid above the liquid, and a folded-back path that communicates with the gas storage chamber and extends to below, and is then

folded-back to extend to above, the bubble generation device releasing the gas passing through the folded-back path as bubbles, the bubble generation device including: a housing that includes a plurality of side plates and a top plate fixed to a plurality of the side plates in an aspect of covering a space surrounded by a plurality of the side plates from above; a plurality of bubble release ports that is disposed in the top plate in an aspect of being arranged along a longitudinal direction of the top plate; and a bubble release chamber that is disposed in the housing and communicates with a plurality of the bubble release ports, wherein each of the gas storage chamber, the folded-back path, and the bubble release chamber is disposed in the housing in an aspect of being arranged in an order of the gas storage chamber, the folded-back path, and the bubble release chamber along a lateral direction of the top plate, the folded-back path is formed by at least a folded-back plate having a cross-sectional shape that extends to below and is then folded back to above, and an inner plate disposed inside the folded-back plate, one end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to one of the two side plates facing each other along the longitudinal direction among a plurality of the side plates, an other end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to the other of the two side plates, one side of the folded-back path in the lateral direction communicates with the bubble release chamber, and an other side of the folded-back path in the lateral direction communicates with the gas storage chamber.

Advantageous Effects of Invention

(11) According to the present invention, there is an excellent effect that a plurality of bubbles can hit the filtration membrane almost simultaneously.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram illustrating a schematic configuration of a water treatment facility using a liquid filtration device according to an embodiment.
- (2) FIG. 2 is a perspective diagram illustrating a membrane module and a bubble generation device of the liquid filtration device.
- (3) FIG. 3 is a perspective diagram illustrating a membrane element and a socket pipe of the membrane module.
- (4) FIG. 4 is a perspective diagram illustrating the bubble generation device of the liquid filtration device.
- (5) FIG. 5 is a cross-sectional diagram of the bubble generation device.
- (6) FIG. 6 is a plan diagram of the bubble generation device.
- (7) FIG. 7 is a cross-sectional diagram illustrating the bubble generation device in a non-operating state.
- (8) FIG. 8 is a cross-sectional diagram illustrating the bubble generation device in a state of a first phase in batch treatment.
- (9) FIG. 9 is a cross-sectional diagram illustrating the bubble generation device in a state of a second phase in batch treatment.
- (10) FIG. 10 is a cross-sectional diagram illustrating the bubble generation device in a state of a third phase in batch treatment.
- (11) FIG. 11 is a cross-sectional diagram illustrating the bubble generation device in a state of a fourth phase in batch treatment.
- (12) FIG. 12 is a perspective diagram illustrating the bubble generation device according to an example.
- (13) FIG. 13 is a cross-sectional diagram illustrating the bubble generation device.
- (14) FIG. 14 is a cross-sectional diagram illustrating an example of a state of the bubble generation

device in the second phase in batch treatment.

(15) FIG. 15 is a cross-sectional diagram illustrating the bubble generation device according to a modification.

(16) FIG. 16 is a perspective diagram illustrating a folded-back plate and an inner plate of a first folded-back path, and a partition plate.

DESCRIPTION OF EMBODIMENTS

(17) Hereinafter, an embodiment of a liquid filtration device to which the present invention is applied will be described with reference to the drawings. In the embodiment, for the sake of easy description, structures and elements other than a main part of the present invention will be described in a simplified or omitted manner. In addition, in the drawings, the same elements are denoted by the same reference numerals. Note that the shapes, dimensions, and the like of the elements illustrated in the drawings are schematically illustrated, and do not indicate actual shapes, dimensions, and the like.

(18) FIG. 1 is a diagram illustrating a schematic configuration of a water treatment facility using a liquid filtration device according to an embodiment. The water treatment facility includes a raw water tank 1, a filtration treatment water tank 2, a treatment water tank 3, a control device 4, a raw water pump 5, a first water level sensor 6, a raw water transfer pipe 7, a treatment water transfer pipe 8, a suction pump 9, a second water level sensor 11, a third water level sensor 12, and the like. In addition, the water treatment facility includes a blower 13, an air supply pipe 14, a mount 15, a membrane module 20, a bubble generation device 30, and the like.

(19) In the raw water tank 1, raw water (pre-treatment water) W.sub.1 as a liquid is stored. The first water level sensor including an ultrasonic sensor or the like installed in the raw water tank 1 senses the water level (water surface height) of the raw water W.sub.1 in the raw water tank 1, and transmits the sensing result to the control device 4 as a water level signal. The raw water pump 5 installed in the raw water tank 1 sucks and discharges the raw water W.sub.1 in the raw water tank 1, and sends the raw water W.sub.1 to the filtration treatment water tank 2 through the raw water transfer pipe 7. As the raw water pump 5, one including a submersible pump has been exemplified, but one including a land pump may be used.

(20) The filtration treatment water tank 2 is a water tank made of reinforced concrete. A liquid filtration device 50 is installed in the filtration treatment water tank 2. The liquid filtration device 50 includes the mount 15, the membrane module 20, and the bubble generation device 30, and is entirely immersed in the raw water W.sub.1 in the filtration treatment water tank 2. The blower 13 discharges the air as the gas sucked from a suction port to the air supply pipe 14 through a discharge port. The air discharged to the air supply pipe 14 is supplied to the bubble generation device 30 of the liquid filtration device 50. The third water level sensor 12 installed in the filtration treatment water tank 2 senses the water level of the raw water W.sub.1 in the filtration treatment water tank 2, and transmits the sensing result to the control device 4 as a water level signal.

(21) The suction pump 9 sucks the raw water W.sub.1 in the filtration treatment water tank 2 through the treatment water transfer pipe 8 and a membrane element described below installed in the membrane module 20. The sucked raw water W.sub.1 is filtrated by the membrane element to become treated water W.sub.2, and then sent to the treatment water tank 3 through the treatment water transfer pipe 8. The second water level sensor 11 set on the treatment water tank 3 senses the water level of the treated water W.sub.2 in the treatment water tank 3, and transmits the sensing result to the control device 4 as a water level signal.

(22) Note that, instead of the suction pump 9, a pump that generates a suction force using hydraulic head pressure may be used. A means for suction is not particularly limited.

(23) When the water level of the treatment water tank 3 does not reach the upper limit and a predetermined operation execution condition is satisfied, the control device 4 operates the suction pump 9 and the blower 13 to execute the filtration treatment of the raw water W.sub.1. However, even when the operation execution condition is satisfied, when the water level of the raw water

W.sub.1 in the raw water tank **1** is lower than or equal to the lower limit and when the water level of the raw water W.sub.1 in the filtration treatment water tank **2** is lower than or equal to the lower limit, the control device **4** stops the execution of the filtration treatment. Note that the role of the blower **13** will be described below.

(24) FIG. **2** is a perspective diagram illustrating the membrane module **20** and the bubble generation device **30** of the liquid filtration device (**50**) according to the embodiment. In the drawing, Z represents a gravity direction. Hereinafter, the configuration of the liquid filtration device (**50**) will be described with reference to the drawings, and when the term “up-down direction” is used, the term “up-down direction” means a direction in which the membrane module **20** and the bubble generation device **30** face each other regardless of the orientations of the membrane module **20** and the bubble generation device **30** illustrated in the drawings. In addition, when the term “above” is used, the term “above” means a direction toward the upper side along the “up-down direction” regardless of the orientations of the membrane module **20** and the bubble generation device **30** illustrated in the drawings. In addition, when the term “below” is used, the term “below” means a direction toward the lower side along the “up-down direction” regardless of the orientations of the membrane module **20** and the bubble generation device **30** illustrated in the drawings. In addition, when the term “horizontal direction” is used, the term “horizontal direction” means a direction orthogonal to the “up-down direction” regardless of the orientations of the membrane module **20** and the bubble generation device **30** illustrated in the drawings.

(25) The membrane module **20** includes a housing **21** having a rectangular tube shape and a plurality of membrane elements **22**. The plurality of membrane elements **22** has a plate shape, is disposed in an aspect of being arranged in a straight line at intervals along the “horizontal direction” in the housing **21**, and is held by the inner surface of the housing **21**.

(26) The bubble generation device **30** having a box shape is disposed “below” (immediately below) the membrane module **20**. The bubble generation device **30** includes two pipe connection portions **31** on a side plate, and the above-described air supply pipe (**14** in FIG. **1**) is connected to each of the pipe connection portions **31**.

(27) FIG. **3** is a perspective diagram illustrating the membrane element **22** and a socket pipe **16**. The membrane element **22** includes a filtration membrane **23** and four holding members **24**. In the liquid filtration device (**50**) according to the embodiment, a filtration membrane made of a flat membrane having a plate shape is used as the filtration membrane **23**, but the type of filtration membrane **23** is not limited to the flat membrane, and other types such as a hollow fiber membrane may be used. The material of the filtration membrane **23** may be an organic material such as polyvinyl chloride (PVC) or polyvinylidene fluoride (PVDF), or may be a ceramic composed of one type or a plurality of types of alumina, cordierite, silicon carbide, and other metal oxides. In addition, the filtration membrane **23** may be a composite membrane in which an organic membrane and a ceramic membrane are combined.

(28) The filtration membrane **23** includes a plurality of hollows **23a** extending in the up-down direction. These hollows **23a** are arranged at predetermined intervals in a lateral direction of the filtration membrane **23**, and the upper ends of the hollows **23a** are openings facing the “above”.

(29) The socket pipe **16** is attached to an upper end portion of the filtration membrane **23**. In addition, the above-described treatment water transfer pipe (**8** in FIG. **1**) is connected to the socket pipe **16**. When the suction pump (**9** in FIG. **1**) is operated, a suction force is generated on the surface of the filtration membrane **23**, and the raw water (W.sub.1 in FIG. **1**) existing around the filtration membrane **23** is sucked into the hollows **23a** through the countless fine holes of the filtration membrane **23**. At this time, the raw water W.sub.1 is filtrated, and the contamination substances in the raw water W.sub.1 remain on the surface of the filtration membrane **23**.

(30) Each of the four holding members **24** is engaged with an engagement portion provided on the inner surface of the housing (**21** in FIG. **2**) in a state of being fixed near the corner of the filtration membrane **23**.

(31) The bubble generation device **30** installed “below” the membrane module **20** releases the bubbles toward the “above” membrane module **20**. The released bubbles float in the raw water W.sub.1 and enter the housing **21** through the opening at the lower end of the housing **21** of the membrane module **20**. Thereafter, the bubbles reach the lower ends of the plurality of filtration membranes **23**, are divided into a plurality of parts by the filtration membranes **23**, and then enter a region between the filtration membranes **23** adjacent to each other (hereinafter, referred to as an “inter-membrane region”). The bubbles floating in the “inter-membrane region” vigorously shake the raw water W.sub.1 near the surface of the filtration membranes **23** to remove the contamination substances adhering to the surfaces of the filtration membranes **23**.

(32) Note that the shape of the housing **21** is a shape surrounding the four sides of the plurality of membrane elements **22** as illustrated in the drawing, but may be a shape having an opening in a part of a side surface of the housing **21** as long as the bubbles do not largely leak from the inside of the housing **21**.

(33) FIG. **4** is a perspective diagram illustrating the bubble generation device **30**. The bubble generation device **30** includes a housing **40**. The housing **40** includes a first side plate **41**, a second side plate **42**, a third side plate **43**, a fourth side plate **44**, and a top plate **32**. The top plate **32** is formed with an elongated hole having a rectangular shape extending along a longitudinal direction of the top plate **32**. A cover plate **32b** having a rectangular shape is fixed to the top plate **32** so as to cover the elongated hole. The cover plate **32b** is a part of the top plate **32**. In the cover plate **32b**, six bubble release ports **32a** having a perfect circular shape are formed in an aspect of being arranged in a straight line. The bubble generation device **30** does not include a bottom plate, and includes a lower end opening **39** that is largely opened to the “below” at a lower end of the bubble generation device **30**.

(34) FIG. **5** is a cross-sectional diagram of the bubble generation device **30**. In addition, FIG. **6** is a plan diagram of the bubble generation device **30**. The bubble generation device **30** includes a first gas storage chamber **33**, a second gas storage chamber **34**, a bubble release chamber **35**, a first folded-back path **36**, a second folded-back path **37**, and the like. Each of the first gas storage chamber **33**, the second gas storage chamber **34**, the bubble release chamber **35**, the first folded-back path **36**, and the second folded-back path is disposed in the housing **40**.

(35) The dashed-dotted line in FIG. **5** is a center line Lc of the top plate **32**. Each of the six bubble release ports **32a** that releases the bubbles including air as a gas is disposed in an aspect that the center thereof is positioned on the center line Lc of the top plate **32** and is opened to the “above”. The bubble release chamber **35** is disposed “below” (immediately below) the six bubble release ports **32a**, and communicates with each of the six bubble release ports **32a**.

(36) As illustrated in FIG. **6**, the shape of the bubble release chamber **35** is an elongated shape extending along the arrangement direction of the six bubble release ports **32a**.

(37) Each of the first gas storage chamber **33** and the second gas storage chamber **34** stores the raw water (treatment target water) as a liquid therein, and stores air as a gas supplied into the raw water “above” the raw water. The first folded-back path **36** communicates with the first gas storage chamber **33** and extends to the “below”, and is then folded back to extend to the “above”. The second folded-back path **37** communicates with the second gas storage chamber **34** and extends to the “below”, and is then folded back to extend to the “above”.

(38) A left-right direction in FIG. **5** is along the lateral direction of the top plate **32**. In addition, a direction orthogonal to the paper surface of FIG. **5** is along the longitudinal direction of the top plate **32**. Each of the first gas storage chamber **33**, the first folded-back path **36**, the bubble release chamber **35**, the second folded-back path **37**, and the second gas storage chamber **34** is disposed in an aspect of being arranged in an order of the first gas storage chamber **33**, the first folded-back path **36**, the bubble release chamber **35**, the second folded-back path **37**, and the second gas storage chamber **34** in the lateral direction of the top plate **32**.

(39) The first folded-back path **36** is formed by at least a folded-back plate **45** having a cross-

sectional shape (more specifically, a U-shaped cross-sectional shape) that extends to the “below” and is then folded back to the “above”, and an inner plate **46** disposed inside the folded-back plate **45**. One end of each of the folded-back plate **45** and the inner plate **46** in the longitudinal direction (the same as the longitudinal direction of the top plate **32**) is fixed to the first side plate **41** as one of the first side plate **41** and the third side plate **43** facing each other along the longitudinal direction among the four side plates (**41** to **44**). The other end of each of the folded-back plate **45** and the inner plate **46** in the longitudinal direction is fixed to the third side plate **43**, which is the other side plate. One side of the first folded-back path **36** in the lateral direction communicates with the bubble release chamber **35**. The other side of the first folded-back path **36** in the lateral direction communicates with the first gas storage chamber **33**.

(40) The folded-back plate **45** of the first folded-back path **36** is fixed to the first side plate **41** and the third side plate **43** in an aspect in which a gap is formed between the top plate **32** and an upper end on one side in the lateral direction while a gap is formed between the top plate **32** and an upper end on the other side in the lateral direction. The first folded-back path **36** and the bubble release chamber **35** communicate with each other through a second communication port **36e** formed of the gap between the upper end on the one side in the lateral direction of the folded-back plate **45** of the first folded-back path **36** and the top plate **32**. In addition, the first folded-back path **36** and the first gas storage chamber **33** communicate with each other through a first communication port **36d**, which is the gap between the upper end on the other side in the lateral direction of the folded-back plate **45** of the first folded-back path **36** and the top plate **32**. The inner plate **46** of the first folded-back path **36** is fixed to the first side plate **41**, the second side plate **42**, and the top plate **32**.

(41) The second folded-back path **37** is formed by at least a folded-back plate **47** having a cross-sectional shape (more specifically, a U-shaped cross-sectional shape) that extends to the “below” and is then folded back to the “above”, and an inner plate **48** disposed inside the folded-back plate **47**. One end of each of the folded-back plate **47** and the inner plate **48** of the second folded-back path **37** in the longitudinal direction is fixed to the first side plate **41**. In addition, the other end of each of the folded-back plate **47** and the inner plate **48** of the second folded-back path **37** in the longitudinal direction is fixed to the third side plate **43**. The other side of the second folded-back path **37** in the lateral direction communicates with the bubble release chamber **35**. In addition, one side of the second folded-back path **37** in the lateral direction communicates with the second gas storage chamber **34**.

(42) The folded-back plate **47** of the second folded-back path **37** is fixed to the first side plate **41** and the third side plate **43** in an aspect in which a gap is formed between the top plate **32** and an upper end on one side in the lateral direction while a gap is formed between the top plate **32** and an upper end on the other side in the lateral direction. The second folded-back path **37** and the bubble release chamber **35** communicate with each other through a first communication port **37e**, which is a gap between the upper end on the other side in the lateral direction of the folded-back plate **47** of the second folded-back path **37** and the top plate **32**. In addition, the second folded-back path **37** and the second gas storage chamber **34** communicate with each other through a second communication port **37d**, which is a gap between the upper end on the one side in the lateral direction of the folded-back plate **47** of the second folded-back path **37** and the top plate **32**. The inner plate **48** of the second folded-back path **37** is fixed to the first side plate **41**, the third side plate **43**, and the top plate **32**.

(43) The structure of the second folded-back path **37** is a structure that is line-symmetric with respect to the center line **Lc** with respect to the first folded-back path **36**. The first gas storage chamber **33** is adjacent to the first folded-back path **36** on the other side in the lateral direction. In addition, the second gas storage chamber **34** is adjacent to the second folded-back path **37** on the one side in the lateral direction. The structure of the second gas storage chamber **34** is a structure that is line-symmetric with respect to the center line **Lc** with respect to the first gas storage chamber **33**.

(44) The first folded-back path **36** includes a descending portion **36a** extending from the first communication port **36d** toward the “below”, a folded-back portion **36b** folded back toward the “above”, and an ascending portion **36c** extending toward the “above” and communicating with the second communication port **36e**. The second folded-back path **37** also includes a similar descending portion **37a**, a similar folded-back portion **37b**, and a similar ascending portion **37c**.

(45) In the bubble generation device **30** having such a configuration, as the air storage amount in the first gas storage chamber **33** increases, the air in the descending portion **36a** of the first folded-back path **36** gradually descends in the descending portion **36a**. When the air storage amount in the first gas storage chamber **33** further increases, the air in the descending portion **36a** reaches the folded-back portion **36b**, ascends at once in the ascending portion **36c**, and enters the bubble release chamber **35**. In addition, as the air storage amount in the second gas storage chamber **34** increases, the air in the descending portion **37a** of the second folded-back path **37** gradually descends in the descending portion **37a**. When the air storage amount in the second gas storage chamber **34** further increases, the air in the descending portion **37a** reaches the folded-back portion **37b**, ascends at once in the ascending portion **37c**, and enters the bubble release chamber **35**.

(46) The air that has ascended in the ascending portion **36c** of the first folded-back path **36** and the air that has ascended in the ascending portion **37c** of the second folded-back path **37** merge in the bubble release chamber **35**, and are then released from the six bubble release ports **32a**. Since substantially uniform pressure is applied to the air in the bubble release chamber **35** in the “horizontal direction” by the raw water existing “below”, the bubbles are released from each of the six bubble release ports **32a** almost simultaneously. Thus, with the bubble generation device **30**, the plurality of bubbles can be caused to hit the filtration membranes **23** almost simultaneously.

(47) FIG. 7 is a cross-sectional diagram illustrating the bubble generation device **30** in a non-operating state. In the drawing, hatching in the cross section of the bubble generation device **30** is omitted for easy viewing. In the bubble generation device **30** in a non-operating state, as illustrated in the drawing, most of the internal space of the bubble generation device **30** is filled with the raw water W.sub.1.

(48) The bubble generation device **30** can intermittently form bubbles from the bubble release ports **32a**. Hereinafter, a process from when the bubbles are released from the bubble release ports **32a** to immediately before the bubbles are released from the bubble release ports **32a** is referred to as batch treatment.

(49) FIG. 8 is a cross-sectional diagram illustrating the bubble generation device **30** in a state of a first phase in batch treatment. One of the two air supply pipes **14** connected to the bubble generation device **30** is opened at a lower portion of the first gas storage chamber **33** and supplies air to the raw water W.sub.1 in the first gas storage chamber **33**. In addition, the other is opened at a lower portion of the second gas storage chamber **34** and supplies air to the raw water W.sub.1 in the second gas storage chamber **34**.

(50) In the first phase of the batch treatment, the air supplied into the raw water W.sub.1 by one of the two air supply pipes **14** floats in the raw water W.sub.1 and is stored in the upper portion of the first gas storage chamber **33**. A part of the air stored in the upper portion of the first gas storage chamber **33** in this manner enters the descending portion **36a** of the first folded-back path **36** through the first communication port **36d**. In the first gas storage chamber **33** and the descending portion **36a** of the first folded-back path **36**, the water level of the raw water W.sub.1 is lowered as the storage amount of air is increased. The water level of the raw water W.sub.1 in the first gas storage chamber **33** is substantially the same as the water level of the raw water W in the first folded-back path **36**.

(51) In the first gas storage chamber **33**, the amount of the raw water W.sub.1 corresponding to the lowering of the water level of the raw water W.sub.1 flows out to the outside of the bubble generation device **30** through the lower end opening (**39** in FIG. 4) of the bubble generation device **30**. In addition, in the descending portion **36a** of the first folded-back path **36**, the amount of the

raw water W.sub.1 corresponding to the lowering of the water level of the raw water W.sub.1 enters the folded-back portion **36b**. As a result of this entry, substantially the same amount of the raw water W.sub.1 ascends in the ascending portion **36c** of the first folded-back path **36**, and enters the bubble release chamber **35** through the second communication port (**36e** in FIG. 5). As a result of this entry, substantially the same amount of the raw water W.sub.1 flows out to the outside of the bubble generation device **30** through the lower end opening or the bubble release ports **32a** of the bubble generation device **30**.

(52) Although the behavior of the air and the raw water W.sub.1 in the first gas storage chamber **33** and the first folded-back path **36** has been described, the same applies to the behavior of the air and the raw water W.sub.1 in the second gas storage chamber **34** and the second folded-back path **37**.

(53) FIG. 9 is a cross-sectional diagram illustrating the bubble generation device **30** in a state of a second phase in batch treatment. In the second phase of the batch treatment, as compared with the first phase, the storage amount of air in each of the first gas storage chamber **33**, the descending portion **36a** of the first folded-back path **36**, the second gas storage chamber **34**, and the descending portion **37a** of the second folded-back path **37** is increased and the water level of the raw water W.sub.1 is lowered.

(54) Immediately before the end of the second phase of the batch treatment, in the first folded-back path **36**, the air in the descending portion **36a** enters the folded-back portion **36b** and passes through the folded-back point of the folded-back portion **36b**.

(55) FIG. 10 is a cross-sectional diagram illustrating the bubble generation device **30** in a state of a third phase in batch treatment. In the third phase, the air that has passed through the folded-back point ascends at a high speed in the ascending portion **36c** of the first folded-back path **36** and reaches the second communication port **36e**, and further exits to the outside of the bubble generation device **30** through the bubble release chamber **35** and the bubble release ports **32a**. In synchronization with such a behavior of air, the air in the first gas storage chamber **33** enters the descending portion **36a** of the first folded-back path **36** through the first communication port **36d** and descends at a high speed in the descending portion **36a**.

(56) Although the behavior of the air in the first gas storage chamber **33** and the first folded-back path **36** has been described, the same applies to the behavior of the air in the second gas storage chamber **34** and the second folded-back path **37**. Above the bubble release ports **32a**, air exiting from the first folded-back path **36** to the outside via the bubble release chamber **35** and air exiting from the second folded-back path **37** to the outside via the bubble release chamber **35** are integrated to form an air aggregate having a circular cross section.

(57) The reason why the large air aggregate as described above is formed above the bubble release ports **32a** is as described below.

(58) That is, it is assumed that the flow of the ascending air toward the bubble release ports **32a** in the bubble release chamber **35** is faster than the flow of the air entering the bubble release chamber **35** from the ascending portions (**36c**, **37c**) through the second communication ports (**36e**, **37e**). In this case, the moving speed of the air in the bubble release chamber **35** is faster than the moving speed of the air flowing into the bubble release chamber **35** from the second communication ports (**36e**, **37e**). Then, a negative pressure is generated in the bubble release chamber **35**. With this negative pressure, the raw water W.sub.1 in the bubble release chamber **35** is drawn into the air, and a gas-liquid mixed flow of the air and the raw water W.sub.1 is generated. As a result, the air that has entered the bubble release chamber **35** from the folded-back paths (**36**, **37**) through the second communication ports (**36e**, **37e**) is divided by the gas-liquid mixed flow, and does not grow into large bubbles.

(59) One of the causes of the moving speed of the air in the bubble release chamber **35** being faster than the moving speed of the air flowing into the bubble release chamber **35** from the second communication ports (**36e**, **37e**) is the flow path resistance of the folded-back portions (**36b**, **37b**).

(60) Therefore, the bubble generation device **30** according to the embodiment includes the lower

end opening (39 in FIG. 4) as an opening facing the “below” at the lower end of each of the two gas storage chambers (33, 34) and the bubble release chamber 35. Since the two gas storage chambers (33, 34) and the bubble release chamber 35 communicate with each other at lower portions, the lower end opening (39 in FIG. 4) functions as a common opening of the two gas storage chambers (33, 34) and the bubble release chamber 35. Strictly speaking, a region immediately below the first gas storage chamber 33 within the entire region of the lower end opening (39 in FIG. 4) is an opening facing “below” at the lower end of the first gas storage chamber 33, and a region immediately below the second gas storage chamber 34 is an opening facing “below” at the lower end of the second gas storage chamber 34. In addition, a region immediately below the bubble release chamber 35 is an opening facing “below” at the lower end of the bubble release chamber 35.

(61) When the air in the bubble release chamber 35 starts to be released to the outside through the bubble release ports 32a, the air existing in the vicinity of the folded-back portions (36b, 37b) starts to move at a speed slightly slower than the preceding air due to the flow path resistance. Then, although a slight negative pressure is generated in the bubble release chamber 35, as indicated by the thick arrows in FIG. 10, the raw water W.sub.1 existing “below” the air applies a force directed to the “above” to the air existing “above”. At this time, immediately below the bubble release chamber 35, the raw water W.sub.1 existing “below” the bubble generation device 30 is caused to enter the inside of the bubble generation device 30 through the lower end opening (39 in FIG. 4), thereby supporting the above-described force. At the same time, in each of the two gas storage chambers (33, 34), the raw water W.sub.1 existing “below” the air applies a force of pushing into the folded-back paths (36, 37) through the first communication ports (36d, 37d) to the air existing “above”. At this time, immediately below each of the two gas storage chambers (33, 34), the raw water W.sub.1 existing “below” the bubble generation device 30 is caused to enter the inside of the bubble generation device 30 through the lower end opening (39 in FIG. 4), thereby supporting the above-described force.

(62) As a result, the air in each of the two folded-back paths (36, 37) smoothly follows the preceding air in the bubble release chamber 35, and quickly flows into the bubble release chamber 35 through the second communication ports (36e, 37e). Therefore, with the bubble generation device 30 according to the embodiment, the succeeding air can be smoothly moved toward the bubble release ports 32a in a state of being continuous with the preceding air without being divided from the preceding air, and can be released from the bubble release ports 32a as bubbles having a large diameter.

(63) As described above, the opening provided at the lower end of the bubble release chamber 35 is a region immediately below the bubble release chamber 35 within the entire region of the lower end opening (39 in FIG. 4). That is, the opening area of the opening provided at the lower end of the bubble release chamber 35 is the same as the cross-sectional area of the bubble release chamber 35 in the horizontal direction. In such a configuration, in the bubble release chamber 35, when bubbles are released, the force for pushing the above air to the above by the raw water W.sub.1 existing below the air can be made uniform without being concentrated on a part in the cross-sectional direction. Thus, it is possible to more reliably suppress drawing of the raw water W.sub.1 existing in the vicinity of the interface with the air by the air and to more reliably increase the diameter of the bubbles.

(64) The opening provided at the lower end of the gas storage chambers (33, 34) is a region immediately below the gas storage chambers (33, 34) within the entire region of the lower end opening (39 in FIG. 4). That is, the opening area of the opening provided at the lower end of the gas storage chambers (33, 34) is the same as the cross-sectional area of the gas storage chambers (33, 34). In such a configuration, when the bubbles are released from the bubble release ports 32a, in the gas storage chambers (33, 34), the force for pushing the above air to the “above” by the raw water W.sub.1 existing below the air can be made uniform without being concentrated on a part in

the cross-sectional direction. Thus, it is possible to suppress the inflow of the raw water W.sub.1 from the gas storage chambers (33, 34) to the folded-back paths (36, 37) by suppressing the drawing of the raw water W.sub.1 existing in the vicinity of the interface with the air by the air.

(65) Note that the shape of the bubble release ports 32a is not limited to a circular shape, and may be another shape such as a rectangular shape. In addition, in the illustrated bubble generation device 30, the cover plate 32b including the six bubble release ports 32a is fixed to the top plate 32, but a plurality of bubble release ports may be provided in the top plate 32 itself. However, in the configuration in which the cover plate 32b is fixed to the top plate 32, the diameter and the disposition interval of the bubble release ports 32a can be easily adjusted by replacing the cover plate 32b.

(66) The bubble generation device 30 according to the embodiment includes the two gas storage chambers (33, 34), but the number of gas storage chambers is not limited to two, and may be one or three or more.

(67) As illustrated in FIG. 4, the bubble release ports 32a are opened to the “above”. In such a configuration, as compared with a configuration including a bubble release port that is opened in a direction different from the “above”, a phenomenon that the “external preceding air” is cut from the “internal succeeding air” by the peripheral wall of the bubble release port is less likely to occur, and thus bubbles having a larger diameter can be formed.

(68) Note that, in the liquid filtration device (50) according to the embodiment, the bubbles released from the six bubble release ports 32a and having mutually equal sizes become one long continuous large bubble, and then the bubble is divided by the plurality of filtration membranes (23) and enter the “inter-membrane region”.

(69) The present inventors experimentally manufactured a test device having the same configuration as the liquid filtration device (50) according to the embodiment, and conducted an experiment of measuring the vibration amount of the filtration membrane (23 in FIG. 3) by scrubbing using bubbles. The fact that the measured value of the vibration amount of the filtration membrane is large suggests that the shake amount of the raw water W.sub.1 in the vicinity of the membrane surface is large and the shear force by scrubbing is high. Therefore, the shearing force of the membrane surface can be indirectly evaluated by measuring the vibration amount.

(70) Next, an example in which a more characteristic configuration is added to the bubble generation device (20) according to the embodiment will be described. Note that the configuration of the bubble generation device (20) according to the example is the same as that of the embodiment unless otherwise noted.

(71) In the bubble generation device 30 according to the embodiment illustrated in FIG. 8, it is assumed that there is a difference in air supply speed between one and the other of the two air supply pipes 14. This may cause a defect described below. That is, in one of the folded-back portion 36b of the first folded-back path 36 and the folded-back portion 37b of the second folded-back path 37, the air inside reaches the folded-back point earlier than that in the other, and the bubble having a small diameter only by the supply of the air is released from the bubble release ports 32a. Thereafter, the air inside the other reaches the folded-back point, and the bubble having a small diameter only by the supply of the air is released from the bubble release ports 32a. Thus, the bubble having a small diameter is released from bubble release ports 32a in a cycle shorter than the cycle in the case of releasing the bubble having a large diameter, which may cause cleaning failure of the surface of the filtration membrane (23).

(72) FIG. 12 is a perspective diagram illustrating the bubble generation device 30 according to an example. In addition, FIG. 13 is a cross-sectional diagram illustrating the bubble generation device 30 according to an example. The bubble generation device 30 according to the example includes a communication pipe 38 that communicates the upper portion (more specifically, the top end) of the first gas storage chamber 33 and the upper portion (more specifically, the top end) of the second gas storage chamber 34.

(73) FIG. 14 is a cross-sectional diagram illustrating an example of a state of the bubble generation device 30 in the second phase in batch treatment. In the illustrated example, the supply speed of air to the second gas storage chamber 34 is higher than the supply speed of air to the first gas storage chamber 33. Nevertheless, the water level of the raw water W.sub.1 is substantially the same in each of the first gas storage chamber 33, the descending portion 36a of the first folded-back path 36, and the descending portion 37a of the second folded-back path 37.

(74) In the second gas storage chamber 34, the storage amount of air is instantaneously increased as compared with the first gas storage chamber 33 due to a difference in air supply amount, but, due to this increase, the air pressure in the second gas storage chamber 34 is higher than the air pressure in the first gas storage chamber 33. Then, the air in the second gas storage chamber 34 moves to the first gas storage chamber 33 on the low pressure side to eliminate the air pressure difference. As a result, the water level of the raw water W.sub.1 is adjusted to the same height in each of the first gas storage chamber 33, the descending portion 36a of the first folded-back path 36, and the descending portion 37a of the second folded-back path 37.

(75) With the bubble generation device 30 having such a configuration, it is possible to avoid cleaning failure of the surface of the filtration membrane (23) due to a difference in air supply speed between the first gas storage chamber 33 and the second gas storage chamber 34.

(76) Note that although the bubble generation device 30 according to the example has a configuration in which the communication pipe 38 can be disassembled by four joints, the communication pipe 38 may have any structure as long as the upper portion of the first gas storage chamber 33 and the upper portion of the second gas storage chamber 34 communicate with each other. In addition, in FIG. 13, the structure of the communication pipe 38 is illustrated in a simplified manner for the sake of convenience.

(77) Next, a modification in which a part of the configuration of the bubble generation device 30 according to the embodiment is changed to another configuration will be described.

(78) FIG. 15 is a cross-sectional diagram illustrating the bubble generation device 30 according to a modification. The bubble generation device 30 according to the modification does not include the second folded-back path. The bubble release chamber and the second gas storage chamber 34 are partitioned by a partition plate 49 fixed to the first side plate 41, the third side plate 43, and the top plate 32. Each of the first gas storage chamber 33, the first folded-back path 36, the bubble release chamber 35, and the second gas storage chamber 34 is disposed in an aspect of being arranged in an order of the first gas storage chamber 33, the first folded-back path 36, the bubble release chamber 35, and the second gas storage chamber 34 in the lateral direction of the top plate 32. The bubble release chamber 35 and the second gas storage chamber 34 communicate with each other via the communication pipe 38.

(79) FIG. 16 is a perspective diagram illustrating the folded-back plate 45 and the inner plate 46 of the first folded-back path 36, and the partition plate 49. A plurality of communication pipes 38 is disposed in an aspect of being arranged along the longitudinal direction of the top plate (32). Each of the plurality of communication pipes 38 penetrates the partition plate 49 and the inner plate 46 to communicate the first gas storage chamber 33 and the second gas storage chamber 34. In addition, each of the plurality of communication pipes 38 is disposed at a position shifted from a position immediately below the bubble release port (32a).

(80) The present inventors have confirmed through experiments that the bubble generation device 30 according to the modification can widen the range of the appropriate air feed amount as compared with the embodiment. The air feed amount is an air supply amount per unit time with respect to the first gas storage chamber 33 and the second gas storage chamber. In addition, the appropriate air feed amount is a range of the air feed amount at which the bubbles can be released at a stable time interval. In the embodiment, the air entering the bubble release chamber 35 from the first folded-back path 36 and the air entering the bubble release chamber 35 from the second folded-back path 37 collide with each other in the bubble release chamber 35. On the other hand,

since the bubble generation device **30** according to the modification does not generate the collision as described above, it is considered that the range of the appropriate air feed amount can be widened as compared with the embodiment.

(81) The present invention is not limited to the above-described embodiment, example, and modification, and a configuration different from the embodiment and the example can be adopted within a range where the configuration of the present invention can be applied. The present invention has unique operations and effects for each aspect described below.

(82) [First Aspect]

(83) A first aspect is a bubble generation device (for example, bubble generation device **30**) including a gas storage chamber (for example, first gas storage chamber) that stores a liquid therein and stores a gas supplied into the liquid above the liquid, and a folded-back path (for example, first folded-back path **36**) that communicates with the gas storage chamber and extends to below, and is then folded back to extend to above, the bubble generation device releasing the gas passing through the folded-back path as bubbles, the bubble generation device including: a housing (for example, housing **40**) that includes a plurality of side plates (for example, first side plate **41** to fourth side plate **44**) and a top plate (for example, top plate **32**) fixed to a plurality of the side plates in an aspect of covering a space surrounded by a plurality of the side plates from above; a plurality of bubble release ports (for example, bubble release ports **32a**) that is disposed in the top plate in an aspect of being arranged along a longitudinal direction of the top plate; and a bubble release chamber (for example, bubble release chamber **35**) that is disposed in the housing and communicates with a plurality of the bubble release ports, wherein each of the gas storage chamber, the folded-back path, and the bubble release chamber is disposed in the housing in an aspect of being arranged in an order of the gas storage chamber, the folded-back path, and the bubble release chamber along a lateral direction of the top plate, the folded-back path is formed by at least a folded-back plate (for example, folded-back plate **45**) having a cross-sectional shape that extends to below and is then folded back to above, and an inner plate (for example, inner plate **46**) disposed inside the folded-back plate, one end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to one (for example, first side plate **41**) of the two side plates facing each other along the longitudinal direction among a plurality of the side plates, an other end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to the other (for example, third side plate **43**) of the two side plates, one side of the folded-back path in the lateral direction communicates with the bubble release chamber, and an other side of the folded-back path in the lateral direction communicates with the gas storage chamber.

(84) In such a configuration, the air entering the bubble release chamber from the folded-back path is pressurized toward the top plate with a uniform pressure, so that the bubbles are released almost simultaneously from each of the plurality of bubble release ports communicating with the bubble release chamber. Thus, according to the first aspect, the plurality of bubbles can be caused to hit the filtration membranes almost simultaneously.

(85) [Second Aspect]

(86) A second aspect includes the configuration of the first aspect, wherein the folded-back plate is fixed, in an aspect of forming a gap between an upper end on one side in the lateral direction and the top plate and forming a gap between an upper end on an other side in the lateral direction and the top plate, to the two top plates, the folded-back path and the bubble release chamber communicate with each other through the gap between the upper end on the one side in the lateral direction of the folded-back plate and the top plate, the folded-back path and the gas storage chamber communicate with each other through the gap between the upper end on the other side in the lateral direction of the folded-back plate and the top plate, and the inner plate is fixed to the two side plates and the top plate.

(87) According to the second aspect, the folded-back path can be formed by combining the folded-back plate and the inner plate, the gas storage chamber and the descending portion of the folded-

back path can communicate with each other, and the ascending portion of the folded-back path and the bubble release chamber can communicate with each other.

(88) [Third Aspect]

(89) A third aspect includes the configuration of the second aspect and includes, in addition to a first gas storage chamber as the gas storage chamber and a first folded-back path as the folded-back path: a second gas storage chamber (for example, second gas storage chamber **34**) and a second folded-back path (for example, second folded-back path **37**), wherein each of the first gas storage chamber, the first folded-back path, the bubble release chamber, the second folded-back path, and the second gas storage chamber is disposed in the lateral direction in an aspect of being arranged in an order of the first gas storage chamber, the first folded-back path, the bubble release chamber, the second folded-back path, and the second gas storage chamber, the second folded-back path is formed by at least a folded-back plate (for example, folded-back plate **47**) having a cross-sectional shape that extends to below and is then folded back to above, and an inner plate (for example, inner plate **48**) disposed inside the folded-back plate, one end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to one of the two side plates facing each other along the longitudinal direction among a plurality of the side plates, an other end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to the other of the two side plates, an other side of the second folded-back path in the lateral direction communicates with the bubble release chamber, and one side of the second folded-back path in the lateral direction communicates with the second gas storage chamber.

(90) According to the third aspect, a gas can be supplied from the first folded-back path and the second folded-back path to the bubble release chamber.

(91) [Fourth Aspect]

(92) A fourth aspect includes the configuration of the third aspect, wherein the folded-back plate of the second folded-back path is fixed, in an aspect of forming a gap between an upper end on one side in the lateral direction and the top plate and forming a gap between an upper end on an other side in the lateral direction and the top plate, to the two top plates, the second folded-back path and the bubble release chamber communicate with each other through the gap between the upper end on the other side in the lateral direction of the folded-back plate of the second folded-back path and the top plate, the second folded-back path and the second gas storage chamber communicate with each other through the gap between the upper end on the one side in the lateral direction of the folded-back plate of the second folded-back path and the top plate, and the inner plate of the second folded-back path is fixed to the two side plates and the top plate.

(93) According to the fourth aspect, the second folded-back path can be formed by combining the folded-back plate and the inner plate, the second gas storage chamber and the descending portion of the second folded-back path can communicate with each other, and the ascending portion of the second folded-back path and the bubble release chamber can communicate with each other.

(94) [Fifth Aspect]

(95) A fifth aspect includes the configuration of the third aspect or the fourth aspect, wherein the first gas storage chamber and the second gas storage chamber communicate with each other via a communication pipe (for example, communication pipe **38**).

(96) According to the fifth aspect, it is possible to avoid cleaning failure of the surface of the filtration membrane due to a difference in gas supply speed between the first gas storage chamber and the second gas storage chamber.

(97) [Sixth Aspect]

(98) A sixth aspect includes the configuration of the second aspect and includes, in addition to a first gas storage chamber as the gas storage chamber and a first folded-back path as the folded-back path: a second gas storage chamber and a second folded-back path, wherein each of the first gas storage chamber, the first folded-back path, the bubble release chamber, the second folded-back path, and the second gas storage chamber is disposed in the lateral direction in an aspect of being

arranged in an order of the first gas storage chamber, the first folded-back path, the bubble release chamber, the second folded-back path, and the second gas storage chamber, the second folded-back path is formed by at least a folded-back plate having a cross-sectional shape that extends to below and is then folded back to above, and an inner plate disposed inside the folded-back plate, one end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to one of the two side plates facing each other along the longitudinal direction among a plurality of the side plates, an other end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to the other of the two side plates, an other side of the second folded-back path in the lateral direction communicates with the bubble release chamber, and one side of the second folded-back path in the lateral direction communicates with the second gas storage chamber.

(99) According to the sixth aspect, the range of the appropriate air feed amount can be widened as compared with the fifth aspect.

(100) [Seventh Aspect]

(101) A seventh aspect is a liquid filtration device (for example, liquid filtration device **50**) including: a filtration membrane (for example, filtration membrane **23**) that includes a membrane portion (for example, a portion of the filtration membrane **23** other than hollows) and a hollow (for example, hollows **23a**) surrounded by the membrane portion, and receives a suction force in the hollow in a state of being disposed in a liquid to take an external liquid into the hollow through the membrane portion and filtrate the liquid; and a bubble generation device that is disposed below the filtration membrane and releases bubbles toward the filtration membrane, wherein the bubble generation device is the bubble generation device according to any of the first to sixth aspects.

(102) With such a configuration, the filtration membrane can be satisfactorily scrubbed by bubbles having a large diameter released from the bubble generation device.

(103) The present application claims priority based on Japanese Patent Application No. 2022-31598 filed on Mar. 2, 2022, the entire contents of which are incorporated herein by reference.

REFERENCE SIGNS LIST

(104) **20** Membrane module **21** Housing **22** Membrane element **23** Filtration membrane **23a** Hollow **30** Bubble generation device **32** Top plate **32a** Bubble release port **33** First gas storage chamber **34** Second gas storage chamber **35** Bubble release chamber **36** First folded-back path **36a** Descending portion **36b** Folded-back portion **36c** Ascending portion **36d** First communication port **36e** Second communication port **36f** Opening **37** Second folded-back path **37a** Descending portion **37b** Folded-back portion **37c** Ascending portion **37d** First communication port **37e** Second communication port **37f** Opening **38** Communication pipe **39** Lower end opening W.sub.1 Raw water (liquid)

Claims

1. A bubble generation device comprising: a housing comprising a plurality of side plates and a top plate fixed to a plurality of the side plates in an aspect of covering a space surrounded by a plurality of the side plates from above, the top plate defining a plurality of bubble release ports arranged along a longitudinal direction of the top plate the housing defining: a gas storage chamber configured to store a liquid therein in a lower portion of the gas storage chamber, the gas storage chamber configured to store a gas supplied into the liquid above the liquid in an upper portion of the gas storage chamber between the lower portion and the top plate; and a bubble release chamber communicating with the plurality of the bubble release ports; a folded-back plate having a cross-sectional shape that extends to the lower portion and is then folded back to the upper portion; an inner plate disposed inside of the folded-back plate, the folded-back plate and the inner plate together forming a folded-back path that communicates with the gas storage chamber and extends to the lower portion, and is folded back to extend to the upper portion, wherein: each of the gas storage chamber, the folded-back path, and the bubble release chamber is disposed in the housing

in an order away from one side plate of the plurality of side plates of (i) the gas storage chamber, (ii) the folded-back path, and (iii) the bubble release chamber along a lateral direction of the top plate that is perpendicular to the longitudinal direction, one end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to one of the two side plates facing each other along the longitudinal direction among a plurality of the side plates, an other end of each of the folded-back plate and the inner plate in the longitudinal direction is fixed to the other of the two side plates, one side of the folded-back path in the lateral direction communicates with the bubble release chamber, an other side of the folded-back path in the lateral direction communicates with the gas storage chamber, and the bubble generation device is configured to release the gas passing through the folded-back path as bubbles.

2. The bubble generation device according to claim 1, wherein: the folded-back plate is fixed to the two side plates such that a first upper end of the folded-back plate at a first lateral side of the folded-back plate forms a first gap between the first upper end and the top plate, and a second upper end of the folded-back plate at a second lateral side of the folded-back plate forms a second gap between the second upper end and the top plate the folded-back path and the bubble release chamber communicate with each other through the second gap, the folded-back path and the gas storage chamber communicate with each other through the first gap, and the inner plate is fixed to the two side plates and the top plate.

3. The bubble generation device according to claim 2, wherein the gas storage chamber is a first gas storage chamber, wherein the folded-back path is a first folded-back path, wherein the bubble generation device further comprises: a second gas storage chamber and a second folded-back path, wherein: each of the first gas storage chamber, the first folded-back path, the bubble release chamber, the second folded-back path, and the second gas storage chamber is disposed in the housing in the order away from the one side plate of (i) the first gas storage chamber, (ii) the first folded-back path, (iii) the bubble release chamber, (iv) the second folded-back path, and (v) the second gas storage chamber, the second folded-back path is formed by at least a second folded-back plate having a cross-sectional shape that extends to the lower portion and is then folded back to the upper portion, and a second inner plate disposed inside the second folded-back plate, one end of each of the second folded-back plate and the second inner plate in the longitudinal direction is fixed to one of the two side plates facing each other along the longitudinal direction among a plurality of the side plates, an other end of each of the second folded-back plate and the second inner plate in the longitudinal direction is fixed to the other of the two side plates, an other side of the second folded-back path in the lateral direction communicates with the bubble release chamber, and one side of the second folded-back path in the lateral direction communicates with the second gas storage chamber.

4. The bubble generation device according to claim 3, wherein: the second folded-back plate of the second folded-back path is fixed to the two side plates such that a first upper end of the second folded-back plate at a first lateral side of the second folded-back plate forms a third gap between the first upper end of the second folded-back plate and the top plate, and a second upper end of the second folded-back plate at a second lateral side of the second folded-back plate forms a fourth gap between the second upper end of the second folded-back plate and the top plate, the second folded-back path and the bubble release chamber communicate with each other through the fourth gap, the second folded-back path and the second gas storage chamber communicate with each other through the third gap, and the second inner plate of the second folded-back path is fixed to the two side plates and the top plate.

5. The bubble generation device according to claim 4, wherein the first gas storage chamber and the second gas storage chamber communicate with each other via a communication pipe.

6. A liquid filtration device comprising: a filtration membrane that includes a membrane portion and a hollow surrounded by the membrane portion, the membrane portion configured to filter an external liquid that is pulled through the membrane portion and into the hollow responsive to a

suction force; and the bubble generation device according to claim 5, the bubble generation device disposed below the filtration membrane and configured to release bubbles toward the filtration membrane.

7. A liquid filtration device comprising: a filtration membrane that includes a membrane portion and a hollow surrounded by the membrane portion, the membrane portion configured to filter an external liquid that is pulled through the membrane portion and into the hollow responsive to a suction force; and the bubble generation device according to claim 4, the bubble generation device disposed below the filtration membrane and configured to release bubbles toward the filtration membrane.

8. The bubble generation device according to claim 3, wherein the first gas storage chamber and the second gas storage chamber communicate with each other via a communication pipe.

9. A liquid filtration device comprising: a filtration membrane that includes a membrane portion and a hollow surrounded by the membrane portion, the membrane portion configured to filter an external liquid that is pulled through the membrane portion and into the hollow responsive to a suction force; and the bubble generation device according to claim 8, the bubble generation device disposed below the filtration membrane and configured to release bubbles toward the filtration membrane.

10. A liquid filtration device comprising: a filtration membrane that includes a membrane portion and a hollow surrounded by the membrane portion, the membrane portion configured to filter an external liquid that is pulled through the membrane portion and into the hollow responsive to a suction force; and the bubble generation device according to claim 3, the bubble generation device disposed below the filtration membrane and configured to release bubbles toward the filtration membrane.

11. The bubble generation device according to claim 2, wherein the gas storage chamber is a first gas storage chamber, wherein the bubble generation device further comprises: a second gas storage chamber, wherein: each of the first gas storage chamber, the folded-back path, the bubble release chamber, and the second gas storage chamber is disposed in the housing in the order away from the one side plate of (i) the first gas storage chamber, (ii) the folded-back path, (iii) the bubble release chamber, and (iv) the second gas storage chamber in the lateral direction, the bubble release chamber and the second gas storage chamber are partitioned by a partition plate fixed to the two side plates and the top plate, and the first gas storage chamber and the second gas storage chamber communicate with each other via a communication pipe.

12. A liquid filtration device comprising: a filtration membrane that includes a membrane portion and a hollow surrounded by the membrane portion, the membrane portion configured to filter an external liquid that is pulled through the membrane portion and into the hollow responsive to a suction force; and the bubble generation device according to claim 11, the bubble generation device disposed below the filtration membrane and configured to release bubbles toward the filtration membrane.

13. A liquid filtration device comprising: a filtration membrane that includes a membrane portion and a hollow surrounded by the membrane portion, the membrane portion configured to filter an external liquid that is pulled through the membrane portion and into the hollow responsive to a suction force; and the bubble generation device according to claim 2, the bubble generation device disposed below the filtration membrane and configured to release bubbles toward the filtration membrane.

14. A liquid filtration device comprising: a filtration membrane that includes a membrane portion and a hollow surrounded by the membrane portion, the membrane portion configured to filter an external liquid that is pulled through the membrane portion and into the hollow responsive to a suction force; and the bubble generation device according to claim 1, the bubble generation device disposed below the filtration membrane and configured to release bubbles toward the filtration membrane.

